

# Newman-Penrose Formalism Calculations

Stephani et al., Equation (wahlquist-newman)

## Metric

$$ds^2 = \left( \frac{U_1(z) - V_1(y)}{v_1(z) + v_2(y)} \right) d\tau \otimes d\tau + \left( \frac{V_1(y) v_1(z) + U_1(z) v_2(y)}{v_1(z) + v_2(y)} \right) d\tau \otimes d\sigma + \left( \frac{V_1(y) v_1(z) + U_1(z) v_2(y)}{v_1(z) + v_2(y)} \right) d\sigma \otimes d\tau + \left( -\frac{V_1(y) v_1(z)^2 - U_1(z) v_2(y)^2}{v_1(z) + v_2(y)} \right) d\sigma \otimes d\sigma + \left( \frac{v_1(z) + v_2(y)}{V_1(y)} \right) dy \otimes dy + \left( \frac{v_1(z) + v_2(y)}{U_1(z)} \right) dz \otimes dz$$

## Coordinates

$$(x^\mu) = (\tau, \sigma, y, z)$$

## Metric Functions

$$v_1 = \frac{\cosh(2\beta z)}{2\beta^2} - \frac{1}{2\beta^2}$$

$$v_2 = -\frac{\cos(2\beta y)}{2\beta^2} + \frac{1}{2\beta^2}$$

$$U_1 = -\frac{1}{2} \gamma^2 z^2 \cosh(2\beta z) + 2\alpha_i \gamma z \cosh(2\beta z) - \alpha_i^2 \cosh(2\beta z) - \alpha_r^2 \cosh(2\beta z) + \frac{3\gamma^2 z \sinh(2\beta z)}{4\beta} - \alpha_i^2 + \alpha_r^2 + Q_0 + \frac{a_1 \sinh(2\beta z)}{2\beta} - \frac{\nu_0 \cosh(2\beta z)}{2\beta^2} + \frac{\mu_0 z \sinh(2\beta z)}{2\beta^3} + \frac{\nu_0}{2\beta^2} - \frac{\mu_0 \cosh(2\beta z)}{2\beta^4} + \frac{\mu_0}{2\beta^4}$$

$$V_1 = \frac{1}{2} \gamma^2 y^2 \cos(2\beta y) - 2\alpha_r \gamma y \cos(2\beta y) + \alpha_i^2 \cos(2\beta y) + \alpha_r^2 \cos(2\beta y) - \frac{3\gamma^2 y \sin(2\beta y)}{4\beta} - \alpha_i^2 + \alpha_r^2 + Q_0 + \frac{a_2 \sin(2\beta y)}{2\beta} - \frac{\nu_0 \cos(2\beta y)}{2\beta^2} - \frac{\mu_0 y \sin(2\beta y)}{2\beta^3} + \frac{\nu_0}{2\beta^2} - \frac{\mu_0 \cos(2\beta y)}{2\beta^4} + \frac{\mu_0}{2\beta^4}$$

## Null Tetrad

*Covariant components*

$$l = \left( \frac{1}{2} \sqrt{2} \sqrt{\frac{V_1(y)}{v_1(z) + v_2(y)}} \right) d\tau + \left( -\frac{1}{2} \sqrt{2} \sqrt{\frac{V_1(y)}{v_1(z) + v_2(y)}} v_1(z) \right) d\sigma + \left( \frac{1}{2} \sqrt{2} \sqrt{\frac{v_1(z)}{V_1(y)} + \frac{v_2(y)}{V_1(y)}} \right) dy$$

$$n = \left( \frac{1}{2} \sqrt{2} \sqrt{\frac{V_1(y)}{v_1(z) + v_2(y)}} \right) d\tau + \left( -\frac{1}{2} \sqrt{2} \sqrt{\frac{V_1(y)}{v_1(z) + v_2(y)}} v_1(z) \right) d\sigma + \left( -\frac{1}{2} \sqrt{2} \sqrt{\frac{v_1(z)}{V_1(y)} + \frac{v_2(y)}{V_1(y)}} \right) dy$$

$$m = \left( \frac{1}{2} \sqrt{2} \sqrt{\frac{U_1(z)}{v_1(z) + v_2(y)}} \right) d\tau + \left( \frac{1}{2} \sqrt{2} \sqrt{\frac{U_1(z)}{v_1(z) + v_2(y)}} v_2(y) \right) d\sigma + \left( \frac{1}{2} i \sqrt{2} \sqrt{\frac{v_1(z)}{U_1(z)} + \frac{v_2(y)}{U_1(z)}} \right) dz$$

$$\bar{m} = \left( \frac{1}{2} \sqrt{2} \sqrt{\frac{U_1(z)}{v_1(z) + v_2(y)}} \right) d\tau + \left( \frac{1}{2} \sqrt{2} \sqrt{\frac{U_1(z)}{v_1(z) + v_2(y)}} v_2(y) \right) d\sigma + \left( -\frac{1}{2} i \sqrt{2} \sqrt{\frac{v_1(z)}{U_1(z)} + \frac{v_2(y)}{U_1(z)}} \right) dz$$

*Contravariant components*

$$l = \left( -\frac{\sqrt{v_1(z) + v_2(y)} v_2(y)}{(\sqrt{2} v_1(z) + \sqrt{2} v_2(y)) \sqrt{V_1(y)}} \right) d\tau + \left( \frac{\sqrt{v_1(z) + v_2(y)}}{(\sqrt{2} v_1(z) + \sqrt{2} v_2(y)) \sqrt{V_1(y)}} \right) d\sigma + \left( \frac{\sqrt{v_1(z) + v_2(y)} \sqrt{V_1(y)}}{\sqrt{2} v_1(z) + \sqrt{2} v_2(y)} \right) dy$$

$$n = \left( -\frac{\sqrt{v_1(z) + v_2(y)} v_2(y)}{(\sqrt{2} v_1(z) + \sqrt{2} v_2(y)) \sqrt{V_1(y)}} \right) d\tau + \left( \frac{\sqrt{v_1(z) + v_2(y)}}{(\sqrt{2} v_1(z) + \sqrt{2} v_2(y)) \sqrt{V_1(y)}} \right) d\sigma + \left( -\frac{\sqrt{v_1(z) + v_2(y)} \sqrt{V_1(y)}}{\sqrt{2} v_1(z) + \sqrt{2} v_2(y)} \right) dy$$

$$m = \left( \frac{\sqrt{v_1(z) + v_2(y)} v_1(z)}{(\sqrt{2} v_1(z) + \sqrt{2} v_2(y)) \sqrt{U_1(z)}} \right) d\tau + \left( \frac{\sqrt{v_1(z) + v_2(y)}}{(\sqrt{2} v_1(z) + \sqrt{2} v_2(y)) \sqrt{U_1(z)}} \right) d\sigma + \left( \frac{i \sqrt{v_1(z) + v_2(y)} \sqrt{U_1(z)}}{\sqrt{2} v_1(z) + \sqrt{2} v_2(y)} \right) dz$$

$$\bar{m} = \left( \frac{\sqrt{v_1(z) + v_2(y)} v_1(z)}{(\sqrt{2} v_1(z) + \sqrt{2} v_2(y)) \sqrt{U_1(z)}} \right) d\tau + \left( \frac{\sqrt{v_1(z) + v_2(y)}}{(\sqrt{2} v_1(z) + \sqrt{2} v_2(y)) \sqrt{U_1(z)}} \right) d\sigma + \left( -\frac{i \sqrt{v_1(z) + v_2(y)} \sqrt{U_1(z)}}{\sqrt{2} v_1(z) + \sqrt{2} v_2(y)} \right) dz$$

## Directional Derivatives

$$D = -\frac{\sqrt{v_1(z) + v_2(y)}v_2(y)}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(y))\sqrt{V_1(y)}}\partial_\tau + \frac{\sqrt{v_1(z) + v_2(y)}}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(y))\sqrt{V_1(y)}}\partial_\sigma + \frac{\sqrt{v_1(z) + v_2(y)}\sqrt{V_1(y)}}{\sqrt{2}v_1(z) + \sqrt{2}v_2(y)}\partial_y$$

$$\Delta = -\frac{\sqrt{v_1(z) + v_2(y)}v_2(y)}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(y))\sqrt{V_1(y)}}\partial_\tau + \frac{\sqrt{v_1(z) + v_2(y)}}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(y))\sqrt{V_1(y)}}\partial_\sigma - \frac{\sqrt{v_1(z) + v_2(y)}\sqrt{V_1(y)}}{\sqrt{2}v_1(z) + \sqrt{2}v_2(y)}\partial_y$$

$$\delta = \frac{\sqrt{v_1(z) + v_2(y)}v_1(z)}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(y))\sqrt{U_1(z)}}\partial_\tau + \frac{\sqrt{v_1(z) + v_2(y)}}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(y))\sqrt{U_1(z)}}\partial_\sigma + \frac{i\sqrt{v_1(z) + v_2(y)}\sqrt{U_1(z)}}{\sqrt{2}v_1(z) + \sqrt{2}v_2(y)}\partial_z$$

$$\bar{\delta} = \frac{\sqrt{v_1(z) + v_2(y)}v_1(z)}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(y))\sqrt{U_1(z)}}\partial_\tau + \frac{\sqrt{v_1(z) + v_2(y)}}{(\sqrt{2}v_1(z) + \sqrt{2}v_2(y))\sqrt{U_1(z)}}\partial_\sigma - \frac{i\sqrt{v_1(z) + v_2(y)}\sqrt{U_1(z)}}{\sqrt{2}v_1(z) + \sqrt{2}v_2(y)}\partial_z$$

## Spin Coefficients

$$\rho = -\frac{i\sqrt{2}\sqrt{v_1(z) + v_2(y)}\sqrt{V_1(y)}\frac{\partial}{\partial z}v_1(z)}{4\left(v_1(z)^2 + 2v_1(z)v_2(y) + v_2(y)^2\right)} - \frac{\sqrt{2}\sqrt{v_1(z) + v_2(y)}\sqrt{V_1(y)}\frac{\partial}{\partial y}v_2(y)}{4\left(v_1(z)^2 + 2v_1(z)v_2(y) + v_2(y)^2\right)}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{i\sqrt{2}\sqrt{v_1(z) + v_2(y)}\sqrt{V_1(y)}\frac{\partial}{\partial z}v_1(z)}{4\left(v_1(z)^2 + 2v_1(z)v_2(y) + v_2(y)^2\right)} - \frac{\sqrt{2}\sqrt{v_1(z) + v_2(y)}\sqrt{V_1(y)}\frac{\partial}{\partial y}v_2(y)}{4\left(v_1(z)^2 + 2v_1(z)v_2(y) + v_2(y)^2\right)}$$

$$\lambda = 0$$



## Relation between Spin Coefficients

$$\bar{\tau} + \pi = -\frac{\sqrt{v_1(z) + v_2(y)}\sqrt{U_1(z)}\frac{\partial}{\partial y}v_2(y)}{\sqrt{2}v_1(z)^2 + 2\sqrt{2}v_1(z)v_2(y) + \sqrt{2}v_2(y)^2}$$

$$\bar{\alpha} + \beta = -\frac{\sqrt{v_1(z) + v_2(y)}\sqrt{U_1(z)}\frac{\partial}{\partial y}v_2(y)}{2\left(\sqrt{2}v_1(z)^2 + 2\sqrt{2}v_1(z)v_2(y) + \sqrt{2}v_2(y)^2\right)}$$

## Ricci Tensor Components

$$\begin{aligned}\Phi_{00} = & \frac{V_1(y)\frac{\partial}{\partial z}v_1(z)^2}{8\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)} + \frac{V_1(y)\frac{\partial}{\partial y}v_2(y)^2}{8\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)} \\ & - \frac{V_1(y)v_1(z)\frac{\partial^2}{(\partial y)^2}v_2(y)}{4\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)} - \frac{V_1(y)v_2(y)\frac{\partial^2}{(\partial y)^2}v_2(y)}{4\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)}\end{aligned}$$

$$\Phi_{01} = \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}\frac{\partial^2}{(\partial z)^2}v_1(z)}{8\left(v_1(z)^2 + 2v_1(z)v_2(y) + v_2(y)^2\right)} - \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}\frac{\partial^2}{(\partial y)^2}v_2(y)}{8\left(v_1(z)^2 + 2v_1(z)v_2(y) + v_2(y)^2\right)}$$

$$\begin{aligned}\Phi_{02} = & -\frac{U_1(z)\frac{\partial}{\partial z}v_1(z)^2}{8\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)} + \frac{U_1(z)v_1(z)\frac{\partial^2}{(\partial z)^2}v_1(z)}{4\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)} \\ & + \frac{U_1(z)v_2(y)\frac{\partial^2}{(\partial z)^2}v_1(z)}{4\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)} - \frac{U_1(z)\frac{\partial}{\partial y}v_2(y)^2}{8\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)}\end{aligned}$$

$$\Phi_{10} = \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}\frac{\partial^2}{(\partial z)^2}v_1(z)}{8\left(v_1(z)^2 + 2v_1(z)v_2(y) + v_2(y)^2\right)} - \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}\frac{\partial^2}{(\partial y)^2}v_2(y)}{8\left(v_1(z)^2 + 2v_1(z)v_2(y) + v_2(y)^2\right)}$$

$$\begin{aligned}
\Phi_{11} = & -\frac{v_1(z)^2 \frac{\partial^2}{(\partial z)^2} U_1(z)}{8(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} - \frac{v_1(z) v_2(y) \frac{\partial^2}{(\partial z)^2} U_1(z)}{4(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} \\
& - \frac{v_2(y)^2 \frac{\partial^2}{(\partial z)^2} U_1(z)}{8(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} + \frac{v_1(z)^2 \frac{\partial^2}{(\partial y)^2} V_1(y)}{8(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} \\
& + \frac{v_1(z) v_2(y) \frac{\partial^2}{(\partial y)^2} V_1(y)}{4(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} + \frac{v_2(y)^2 \frac{\partial^2}{(\partial y)^2} V_1(y)}{8(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} \\
& + \frac{v_1(z) \frac{\partial}{\partial z} U_1(z) \frac{\partial}{\partial z} v_1(z)}{4(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} + \frac{v_2(y) \frac{\partial}{\partial z} U_1(z) \frac{\partial}{\partial z} v_1(z)}{4(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} \\
& - \frac{3U_1(z) \frac{\partial}{\partial z} v_1(z)^2}{16(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} + \frac{3V_1(y) \frac{\partial}{\partial z} v_1(z)^2}{16(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} \\
& + \frac{U_1(z) v_1(z) \frac{\partial^2}{(\partial z)^2} v_1(z)}{8(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} + \frac{U_1(z) v_2(y) \frac{\partial^2}{(\partial z)^2} v_1(z)}{8(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} \\
& - \frac{v_1(z) \frac{\partial}{\partial y} V_1(y) \frac{\partial}{\partial y} v_2(y)}{4(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} - \frac{v_2(y) \frac{\partial}{\partial y} V_1(y) \frac{\partial}{\partial y} v_2(y)}{4(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} \\
& - \frac{3U_1(z) \frac{\partial}{\partial y} v_2(y)^2}{16(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} + \frac{3V_1(y) \frac{\partial}{\partial y} v_2(y)^2}{16(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} \\
& - \frac{V_1(y) v_1(z) \frac{\partial^2}{(\partial y)^2} v_2(y)}{8(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} - \frac{V_1(y) v_2(y) \frac{\partial^2}{(\partial y)^2} v_2(y)}{8(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)}
\end{aligned}$$

$$\Phi_{12} = \frac{\sqrt{U_1(z)} \sqrt{V_1(y)} \frac{\partial^2}{(\partial z)^2} v_1(z)}{8(v_1(z)^2 + 2v_1(z) v_2(y) + v_2(y)^2)} - \frac{\sqrt{U_1(z)} \sqrt{V_1(y)} \frac{\partial^2}{(\partial y)^2} v_2(y)}{8(v_1(z)^2 + 2v_1(z) v_2(y) + v_2(y)^2)}$$

$$\begin{aligned}
\Phi_{20} = & -\frac{U_1(z) \frac{\partial}{\partial z} v_1(z)^2}{8(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} + \frac{U_1(z) v_1(z) \frac{\partial^2}{(\partial z)^2} v_1(z)}{4(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} \\
& + \frac{U_1(z) v_2(y) \frac{\partial^2}{(\partial z)^2} v_1(z)}{4(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)} - \frac{U_1(z) \frac{\partial}{\partial y} v_2(y)^2}{8(v_1(z)^3 + 3v_1(z)^2 v_2(y) + 3v_1(z) v_2(y)^2 + v_2(y)^3)}
\end{aligned}$$

$$\Phi_{21} = \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}\frac{\partial^2}{(\partial z)^2}v_1(z)}{8\left(v_1(z)^2 + 2v_1(z)v_2(y) + v_2(y)^2\right)} - \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}\frac{\partial^2}{(\partial y)^2}v_2(y)}{8\left(v_1(z)^2 + 2v_1(z)v_2(y) + v_2(y)^2\right)}$$

$$\Phi_{22} = \frac{V_1(y)\frac{\partial}{\partial z}v_1(z)^2}{8\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)} + \frac{V_1(y)\frac{\partial}{\partial y}v_2(y)^2}{8\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)}$$

$$- \frac{V_1(y)v_1(z)\frac{\partial^2}{(\partial y)^2}v_2(y)}{4\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)} - \frac{V_1(y)v_2(y)\frac{\partial^2}{(\partial y)^2}v_2(y)}{4\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)}$$

$$\Lambda = -\frac{v_1(z)^2\frac{\partial^2}{(\partial z)^2}U_1(z)}{24\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)} - \frac{v_1(z)v_2(y)\frac{\partial^2}{(\partial z)^2}U_1(z)}{12\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)}$$

$$- \frac{v_2(y)^2\frac{\partial^2}{(\partial z)^2}U_1(z)}{24\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)} - \frac{v_1(z)^2\frac{\partial^2}{(\partial y)^2}V_1(y)}{24\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)}$$

$$- \frac{v_1(z)v_2(y)\frac{\partial^2}{(\partial y)^2}V_1(y)}{12\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)} - \frac{v_2(y)^2\frac{\partial^2}{(\partial y)^2}V_1(y)}{24\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)}$$

$$+ \frac{U_1(z)\frac{\partial}{\partial z}v_1(z)^2}{48\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)} + \frac{V_1(y)\frac{\partial}{\partial z}v_1(z)^2}{48\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)}$$

$$- \frac{U_1(z)v_1(z)\frac{\partial^2}{(\partial z)^2}v_1(z)}{24\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)} - \frac{U_1(z)v_2(y)\frac{\partial^2}{(\partial z)^2}v_1(z)}{24\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)}$$

$$+ \frac{U_1(z)\frac{\partial}{\partial y}v_2(y)^2}{48\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)} + \frac{V_1(y)\frac{\partial}{\partial y}v_2(y)^2}{48\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)}$$

$$- \frac{V_1(y)v_1(z)\frac{\partial^2}{(\partial y)^2}v_2(y)}{24\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)} - \frac{V_1(y)v_2(y)\frac{\partial^2}{(\partial y)^2}v_2(y)}{24\left(v_1(z)^3 + 3v_1(z)^2v_2(y) + 3v_1(z)v_2(y)^2 + v_2(y)^3\right)}$$

## Weyl Tensor Components

$$\Psi_0 = 0$$

$$\begin{aligned}\Psi_1 = & \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}\frac{\partial}{\partial z}v_1(z)^2}{8\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} - \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}v_1(z)\frac{\partial^2}{(\partial z)^2}v_1(z)}{8\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} \\ & - \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}v_2(y)\frac{\partial^2}{(\partial z)^2}v_1(z)}{8\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} + \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}\frac{\partial}{\partial y}v_2(y)^2}{8\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} \\ & - \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}v_1(z)\frac{\partial^2}{(\partial y)^2}v_2(y)}{8\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} - \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}v_2(y)\frac{\partial^2}{(\partial y)^2}v_2(y)}{8\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)}\end{aligned}$$

$$\begin{aligned}\Psi_2 = & \frac{v_1(z)^2\frac{\partial^2}{(\partial z)^2}U_1(z)}{12\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} + \frac{v_1(z)v_2(y)\frac{\partial^2}{(\partial z)^2}U_1(z)}{6\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} \\ & + \frac{v_2(y)^2\frac{\partial^2}{(\partial z)^2}U_1(z)}{12\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} + \frac{v_1(z)^2\frac{\partial^2}{(\partial y)^2}V_1(y)}{12\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} \\ & + \frac{v_1(z)v_2(y)\frac{\partial^2}{(\partial y)^2}V_1(y)}{6\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} + \frac{v_2(y)^2\frac{\partial^2}{(\partial y)^2}V_1(y)}{12\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} \\ & - \frac{v_1(z)\frac{\partial}{\partial z}U_1(z)\frac{\partial}{\partial z}v_1(z)}{4\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} - \frac{v_2(y)\frac{\partial}{\partial z}U_1(z)\frac{\partial}{\partial z}v_1(z)}{4\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} \\ & - \frac{i v_1(z)\frac{\partial}{\partial y}V_1(y)\frac{\partial}{\partial z}v_1(z)}{4\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} - \frac{i v_2(y)\frac{\partial}{\partial y}V_1(y)\frac{\partial}{\partial z}v_1(z)}{4\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} \\ & + \frac{U_1(z)\frac{\partial}{\partial z}v_1(z)^2}{3\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} - \frac{V_1(y)\frac{\partial}{\partial z}v_1(z)^2}{6\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} \\ & - \frac{U_1(z)v_1(z)\frac{\partial^2}{(\partial z)^2}v_1(z)}{6\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} - \frac{U_1(z)v_2(y)\frac{\partial^2}{(\partial z)^2}v_1(z)}{6\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} \\ & + \frac{i v_1(z)\frac{\partial}{\partial z}U_1(z)\frac{\partial}{\partial y}v_2(y)}{4\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} + \frac{i v_2(y)\frac{\partial}{\partial z}U_1(z)\frac{\partial}{\partial y}v_2(y)}{4\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} \\ & - \frac{v_1(z)\frac{\partial}{\partial y}V_1(y)\frac{\partial}{\partial y}v_2(y)}{4\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} - \dots\end{aligned}$$



$$\begin{aligned}\Psi_3 = & \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}\frac{\partial}{\partial z}v_1(z)^2}{8\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} - \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}v_1(z)\frac{\partial^2}{(\partial z)^2}v_1(z)}{8\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} \\ & - \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}v_2(y)\frac{\partial^2}{(\partial z)^2}v_1(z)}{8\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} + \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}\frac{\partial}{\partial y}v_2(y)^2}{8\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} \\ & - \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}v_1(z)\frac{\partial^2}{(\partial y)^2}v_2(y)}{8\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)} - \frac{\sqrt{U_1(z)}\sqrt{V_1(y)}v_2(y)\frac{\partial^2}{(\partial y)^2}v_2(y)}{8\left(v_1(z)^3+3v_1(z)^2v_2(y)+3v_1(z)v_2(y)^2+v_2(y)^3\right)}\end{aligned}$$

$$\Psi_4 = 0$$

## Newman-Penrose Equations

*Equations are vanished.*

## Bianchi Identities

*Equations are vanished.*

## Petrov Type

Type "I"